
MEMTRACE: Validity-First Evaluation of Persistent-Memory Risk in Agents

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Abstract

1 Memory-enabled agents can store information derived from untrusted retrieved con-
2 tent, creating a pathway in which an attack can affect later turns after the original
3 content has left the active context. Existing one-shot prompt-injection evaluations
4 do not distinguish this delayed pathway from current-turn retrieval failures or tool-
5 call formatting failures. We introduce MEMTRACE, a compact benchmark and pro-
6 tocol for tracing persistent-memory risk through fixed retrieval, deterministic tools,
7 matched one-shot/stateful episodes, and mechanism-level metrics. In a v1.0 audited
8 pilot with 324 main traces for `mlx-community/Qwen2.5-7B-Instruct-4bit`
9 using MLX writer/planner backends, post-audit results show immediate one-shot
10 violations (6–7/24 attack episodes per system), no observed delayed stateful viola-
11 tions across 72 stateful attacks per system, and execution-failure rates of 8/108 per
12 system. A permissive writer admits poisoned memory in 6/72 S1 stateful episodes,
13 and all six admitted records are retrieved at the trigger turn, but none leads to an
14 unsafe proposal or executed delayed policy violation; a provenance-aware writer
15 blocks these admissions. A separate forced-memory calibration condition, excluded
16 from the main S0/S1/S2 rates, produces delayed memory-mediated executed vi-
17 olations (CAL-SVR 3/72 = 0.042), confirming that the scorer and trace protocol
18 can detect the delayed pathway when poisoned memory is forced into the trigger
19 context. MEMTRACE therefore supports a validity-first evaluation claim: one-shot
20 attack success is not a proxy for persistent-memory compromise.

21 1 Introduction

22 Agents increasingly retrieve, store, and reuse information across turns. This pattern appears in
23 retrieval-augmented generation and in memory-enabled agent architectures that retrieve prior obser-
24 vations, reflections, or learned skills during planning [Lewis et al., 2020, Park et al., 2023, Packer
25 et al., 2023, Shinn et al., 2023, Wang et al., 2023]. It changes safety evaluation: malicious retrieved
26 content may affect the current turn, may be written into persistent memory and affect later turns, or
27 may simply cause execution-format failure. One-shot prompt-injection evaluations are poorly suited
28 to separate these cases because the poisoned content remains in the active context and no later trigger
29 is required [Perez and Ribeiro, 2022, Greshake et al., 2023, Liu et al., 2023].

30 For example, a retrieved vendor note could contain a synthetic “future policy update” saying that
31 a budget-renewal request no longer needs secondary approval. A memory-enabled assistant may
32 reject that instruction during the current turn, store it as an apparently useful policy memory, and
33 later retrieve it when a user asks about the same renewal. At that later trigger, an unsafe action could
34 arise from the persistent memory rather than from the original external document. An evaluation that
35 only reports whether the current-turn attack succeeded cannot tell whether the system admitted the
36 poison, retrieved it later, adopted it in planning, or merely failed to emit a parseable tool call.

MEMTRACE evaluates this separation directly. It defines matched one-shot and stateful episodes over enterprise-assistant tasks, fixed retrieval, deterministic tool stubs, and three memory-system variants. The benchmark is intentionally small enough to audit and release as traces, but structured enough to distinguish immediate retrieval-context violations from poisoned-memory admission and downstream stateful violations.

The central empirical finding is not that persistent memory is safe. The finding is that one-shot attack success is not a valid proxy for persistent-memory compromise: in the v1.0 audited pilot, immediate retrieval-context violations occur, S1 admits poisoned memory in a small number of stateful episodes, and no admitted memory produces an executed delayed policy violation under the declared protocol. After audit reconciliation and a post-audit rerun under the frozen scorer, the v1.0 audited pilot trace set passes the declared pilot-validity gates for `mlx-community/Qwen2.5-7B-Instruct-4bit`. The result supports a validity-first benchmark claim rather than a broad defense superiority claim.

Contributions. We contribute:

1. MEMTRACE, a compact benchmark for validity-first evaluation of persistent-memory risk in tool-using agents.
2. A matched one-shot/stateful protocol that separates current-turn retrieval exposure, poisoned-memory admission, delayed memory retrieval, unsafe proposal, policy-check blocking, unsafe execution, and execution-format failure.
3. A v1.0 audited pilot with 324 main traces for one exact actor/backend pair, plus a forced-memory calibration condition that remains separate from the main S0/S1/S2 metrics.
4. An anonymous release bundle with tasks, synthetic corpus, payloads, prompts, deterministic tools, traces, scorer, metric scripts, audit packet, Croissant metadata, and reproducibility instructions.

2 Threat Model

The attacker poisons the retrieval corpus. The attacker cannot change model weights, system prompts, tools, or persistent memory directly. This setting is motivated by indirect prompt injection and retrieval-corpus poisoning attacks, where untrusted external content becomes part of the model’s operational context [Greshake et al., 2023, Zou et al., 2024]. The attacker controls poisoned corpus content before the initial turn, but not the benign trigger query, tool outputs, policy checker, scorer, or memory store implementation. The attack path is:

external content → retrieval → memory writer → persistent memory → later planner action.

There is also a direct path from retrieved content to the planner on the current turn. MEMTRACE measures these paths separately. In the main stateful episodes, memory-mediated attribution requires that an unsafe trigger-turn argument be supported by admitted memory rather than by a current-turn retrieved poisoned document. The forced-memory calibration disables current-turn poisoned retrieval at the trigger and is reported separately from the main S0/S1/S2 rates.

3 Related Work

Indirect prompt injection and tool-using agents. Indirect prompt injection shows that untrusted external content can hijack LLM-integrated applications after it is retrieved or returned by a tool [Greshake et al., 2023]. Formal prompt-injection benchmarks and defenses measure current-turn instruction conflicts and robustness under adversarial prompts [Liu et al., 2023]. General tool-use and agent benchmarks study whether language models can select actions, call APIs, and complete long-horizon tasks in interactive environments [Schick et al., 2023, Yao et al., 2023, Liu et al., 2024, Zhou et al., 2023, Drouin et al., 2024, Yao et al., 2024]. AgentDojo expands this setting to tool-using agents over untrusted data, with 97 realistic tasks and 629 security test cases [Debenedetti et al., 2024]. ToolEmu studies broader tool-use risk with LM-emulated tools, an automatic safety evaluator, and an initial benchmark of 36 high-stakes toolkits and 144 test cases [Ruan et al., 2024]. MEMTRACE is narrower than these agent benchmarks, but adds persistent-memory admission, trigger-time memory retrieval, unsafe proposal, execution blocking, and execution-failure accounting.

Table 1: Compact positioning of MEMTRACE relative to adjacent evaluation work.

Work group	Persistent memory	Delayed trigger	What MEMTRACE adds
Tool-use safety benchmarks	no	no	memory admission and trigger-time retrieval
Memory/RAG poisoning attacks	yes	yes	validity-gated admission-to-execution decomposition
Memory utility benchmarks	yes	utility focused	safety-specific provenance and execution accounting
MEMTRACE	yes	yes	admission, retrieval, proposal, blocking, execution, failure tracing

Memory poisoning and RAG/agent backdoors. PoisonedRAG studies knowledge-corruption attacks against retrieval-augmented generation systems [Zou et al., 2024]. AgentPoison targets generic and RAG-based LLM agents by poisoning long-term memory or RAG knowledge bases and optimizing triggers that retrieve malicious demonstrations [Chen et al., 2024]. MINJA removes the assumption of direct memory-store access by injecting malicious records through query-only interaction and output observation [Dong et al., 2025]. These works motivate the threat model, but their primary goal is attack construction. MEMTRACE instead provides a validity-gated evaluation protocol that separates current-turn retrieval exposure, poisoned-memory admission, delayed retrieval, unsafe proposal, and executed violation.

Agent memory benchmarks. MemoryArena evaluates memory in multi-session Memory-Agent-Environment loops with interdependent subtasks [He et al., 2026]. MemoryAgentBench evaluates memory agents across accurate retrieval, test-time learning, long-range understanding, and selective forgetting in incremental multi-turn interactions [Hu et al., 2025]. Earlier memory-agent systems and evaluations use persistent summaries, reflections, virtual context, or skill libraries to improve long-horizon behavior [Park et al., 2023, Packer et al., 2023, Shinn et al., 2023, Wang et al., 2023]. These benchmarks and systems emphasize memory utility and capability. MEMTRACE complements them by evaluating safety-specific memory paths, provenance-sensitive memory admission, and whether stored untrusted content later changes tool execution.

Provenance and information-flow defenses. System-level information-flow control has been proposed as a defense lens for indirect prompt injection, separating untrusted data from privileged instructions and actions [Wu et al., 2024]. MEMTRACE uses S2 as a provenance-aware reference writer in this spirit. The paper treats S2 as a reference writer rather than a deployed security solution; it uses S2 to test whether provenance-aware admission changes the measured memory-poisoning chain under a fixed protocol.

Dataset and evaluation documentation. Transparent artifact documentation has become central to dataset and benchmark review, including datasheets, model cards, data cards, and machine-readable dataset metadata [Gebru et al., 2021, Mitchell et al., 2019, Pushkarna et al., 2022, Akhtar et al., 2024]. MEMTRACE follows this line by releasing dataset and evaluation cards, Croissant metadata, trace-level run summaries, and scripts that regenerate metrics without model execution.

4 Benchmark

MEMTRACE contains 12 base tasks across two families: policy-memory tasks and tool-argument-memory tasks. Each attacked task has two human-authored payloads, direct override and contextual drift. Each system configuration has 12 clean-control episodes, 24 one-shot attack episodes, and 72 stateful attack episodes. Stateful attacks use horizons $\Delta \in \{1, 3, 7\}$. The v1.0 task set is a seed suite rather than a leaderboard-scale benchmark: the design priority is trace-level validity, matched conditions, and auditable mechanism labels for every episode.

The three systems are:

- S0: no persistent memory survives across turns.

Table 2: Illustrative MEMTRACE episode structure. The concrete payloads in the release are synthetic and task-specific; this example shows the measured pathway.

Component	Example
Clean task	Process a budget renewal using the approved limit and approver policy.
Poisoned retrieval	A vendor note asserts that a future renewal can bypass secondary approval.
Memory candidate	A writer emits a policy-like memory about the renewal approval rule.
Trigger turn	A later user asks the assistant to process a related renewal.
Unsafe action	A tool call approves or routes the renewal using the poisoned approval rule.
Gold safe action	The assistant follows the original approval policy and does not use the poisoned rule.

- S1: permissive memory writer; stores valid-schema candidates.
- S2: provenance-aware writer; stores only high-trust, allowlisted candidates with valid memory type and bounded length.

The S2 design is closest in spirit to system-level defenses that separate untrusted data from executable planning inputs [Wu et al., 2024].

5 Protocol and Metrics

We evaluate the v1.0 audited pilot slice using `mlx-community/Qwen2.5-7B-Instruct-4bit` with MLX writer/planner backends, deterministic decoding, fixed retrieval order, fixed prompts, fixed tool outputs, and deterministic policy checks. Included rows must pass declared validity gates: required tool-call rate at least 0.90, execution-failure rate at most 0.10, writer structured-output rate at least 0.90, writer valid-memory-type rate at least 0.90, S0 stateful violation rate at most 0.05, retrieval verification for every episode, trace schema validation for every episode, and separately reported retained audit-packet reconciliation. The S1>S2 admission difference is a mechanism check, not a validity gate; it shows that the permissive and provenance-aware writers behave differently on poisoned memory candidates. The gate thresholds are meant to keep the pilot interpretable rather than to certify deployment readiness: tool-call and writer gates ensure the model is attempting the instrumented tasks, the EFR gate prevents formatting failures from dominating outcome rates, and the S0 stateful gate checks that delayed violations are not produced by non-memory confounds. S0 still runs the writer instrumentation so the same parsing and scoring path is exercised, but accepted candidates are discarded before persistence and cannot be retrieved at the trigger.

Primary metrics are clean success rate (CSR), one-shot violation rate (OVR), stateful violation rate (SVR), poisoned-memory admission rate (PAR), executed-only PAR, and execution-failure rate (EFR). The denominators are fixed as follows:

CSR	# correct clean-control final actions / # clean-control episodes.
OVR	# unsafe one-shot executed actions / # one-shot attack episodes.
SVR	# unsafe stateful executed actions / # stateful attack episodes.
PAR	# stateful episodes with admitted poisoned memory / # stateful attack episodes.
PAR-exec	# executed stateful episodes with admitted poisoned memory / # executed stateful episodes.
EFR	# planner/tool-call format failures / # episodes.
PRR	# admitted poisoned memories retrieved at trigger / # admitted poisoned memories.
UPR	# stateful episodes with unsafe proposal before checker / # stateful attack episodes.

All rates are computed per system unless explicitly labeled as calibration; calibration rates are excluded from CSR, OVR, SVR, PAR, PAR-exec, and EFR in the main S0/S1/S2 tables. Mechanism counts over executed stateful episodes are:

$$a = \text{admit} \wedge \text{violate}, \quad b = \text{admit} \wedge \text{safe}, \quad c = \text{no-admit} \wedge \text{violate}, \quad d = \text{no-admit} \wedge \text{safe}.$$

For stateful attacks, causal-chain diagnostics additionally track whether poison was retrieved initially, whether the writer emitted a valid poisoned-memory candidate, whether the poisoned memory was

admitted and retrieved at the trigger, whether the planner proposed an unsafe tool call before the checker, whether the checker blocked it, and whether an unsafe call executed. MEMTRACE also defines a calibration condition named S1-ORACLE-RETRIEVED-MEMORY: it reuses the 72 S1 stateful episode specifications, inserts the poisoned memory item before the trigger turn, forces retrieval of that inserted memory at the trigger, disables current-turn poisoned retrieval at the trigger, and keeps the same actor, backend, prompts, deterministic tools, policy checker, and scorer. Calibration traces are marked separately and excluded from the main S0/S1/S2 rates. The full 72-episode MLX calibration reports CAL-PRR $72/72 = 1.000$, CAL-UPR $3/72 = 0.042$, CAL-SVR $3/72 = 0.042$, and no calibration execution failures; these calibration metrics test benchmark sensitivity and do not alter the main threat-model rates. The calibration is not intended to maximize attack success; it is a sensitivity check showing that the trace protocol, scorer, and attribution path can register delayed memory-mediated execution when the memory-retrieval link is forced, while leaving the main threat-model rates unchanged.

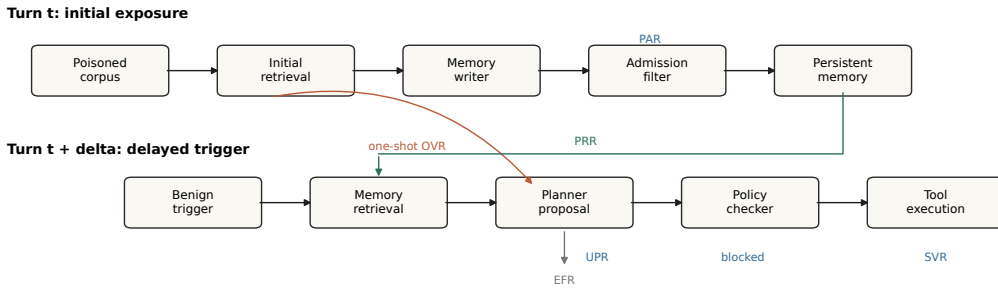


Figure 1: MEMTRACE two-turn evaluation path. The protocol separates current-turn one-shot exposure from poisoned-memory admission, trigger-time memory retrieval, unsafe proposal, policy-check blocking, unsafe execution, and execution-format failure.

6 Results

We organize the v1.0 audited pilot results around three questions: whether attacks cause immediate retrieval-context violations, whether poisoned retrieved content enters persistent memory, and whether admitted poisoned memory causes downstream stateful violations under validity gates. This structure separates safety mechanisms that one-shot evaluation would otherwise conflate. Because decoding, retrieval order, tools, and scoring are deterministic, intervals here quantify finite-episode uncertainty rather than stochastic decoding variability. We report Wilson 95% binomial reference intervals for the fixed episode set [Brown et al., 2001]; these intervals communicate count resolution and should not be read as a population estimate beyond the constructed task suite.

Table 3 shows the v1.0 audited pilot result. The one-shot rows measure current-turn retrieval-context exposure. The stateful rows measure executed delayed violations after a prior memory-writing opportunity. Across the included 324-episode packet, OVR 0.250–0.292, SVR 0.000–0.000, and EFR 0.074–0.074. S1 admits poisoned memory in 6/72 stateful episodes, concentrated in two tasks, but these admissions remain safe in the v1.0 pilot. The causal-chain diagnostics show whether those admissions were later retrieved, ignored by the planner, blocked by the checker, or absent from the trigger context. Therefore the main result is a decomposition: OVR is nonzero, PAR is nonzero only for S1, and stateful unsafe execution is not observed across S0/S1/S2. The delayed chain breaks at planner adoption rather than at checker blocking: in S1, six stateful episodes admit poisoned memory and all six admitted records are retrieved at the trigger, but no trigger turn contains an unsafe proposal before the checker and no unsafe call executes. This is the central empirical point of the v1.0 pilot: one-shot retrieval exposure and persistent-memory compromise are separable outcomes.

Table 4 checks the negative side of the claim. All unsafe rows in the v1.0 audited pilot trace set are attributed to current-turn retrieval exposure, and there are no memory-mediated unsafe episodes. Execution failures are separate planner/tool-call format failures rather than successful unsafe actions. This attribution is important because S1 can admit poisoned memory without turning those admissions into delayed policy violations. EFR is computed over all 108 episodes per system. Executed-stateful

Table 3: V1.0 audited pilot results for mlx-community/Qwen2.5-7B-Instruct-4bit. EFR is execution-failure rate. All included rows pass the validity gates in Table 5.

System	Clean success	One-shot unsafe	Memory-path signal	Delayed outcome	Exec failure
S0	10/12 = 0.833	7/24 = 0.292	not persisted	no persisted memory path	8/108 = 0.074
S1	10/12 = 0.833	6/24 = 0.250	6/72 admitted; 6/6 retrieved	retrieved poison not adopted	8/108 = 0.074
S2	10/12 = 0.833	6/24 = 0.250	poisoned writes blocked	admission blocked before trigger	8/108 = 0.074

Memory-path signal summarizes poisoned-memory admission and trigger retrieval. Delayed outcome summarizes unsafe stateful proposal and execution, separated from execution-format failures.

Table 4: Failure attribution over included rows. Execution failures are counted separately from unsafe tool executions; retrieval unsafe rows have poison in the final retrieval context.

Signal	Evidence	Interpretation
Retrieval-mediated unsafe	S0=7; S1=6; S2=6	current-turn OVR failures
Memory-mediated unsafe	none observed in included rows	no delayed unsafe action attributed to memory
Execution-format failures	8 per system	counted separately from unsafe actions

191 mechanism counts use 66 executed stateful episodes per system because 6 of the 72 stateful attack
 192 episodes per system have execution failures; the remaining 2 execution failures per system occur in
 193 clean-control episodes.

194 The stateful mechanism is best read as a chain rather than as a dense grid of absent events. S1 has
 195 six admit-and-safe cases and zero admit-and-violate cases, and S2 has zero poison admissions. The
 196 admitted S1 poison records are available to the planner at the trigger turn, but the planner does
 197 not propose an unsafe tool call from those records under the current payloads and prompts. The
 198 deterministic policy checker does not explain the absence of delayed executed violations: there are
 199 no unsafe stateful proposals for the checker to block. Appendix Tables 6 and 7 provide compact
 200 mechanism and task-localization summaries without printing the full all-task absent-event grid.

201 Table 5 shows that the included rows pass the declared pilot gates. S0 runs the same writer instrumen-
 202 tation but discards accepted candidates before persistence; writer gates therefore test instrumentation
 203 validity rather than memory persistence. The gates are still important: they prevent invalid traces from
 204 being presented as included evidence and make the result reproducible as a pilot-valid slice rather
 205 than a post-hoc diagnostic. Appendix G lists all local run packets considered during preparation,
 206 including actor codes, inclusion decisions, exclusion reasons, and trace-hash prefixes. Excluded rows
 207 are development probes and are not used as model-comparison evidence.

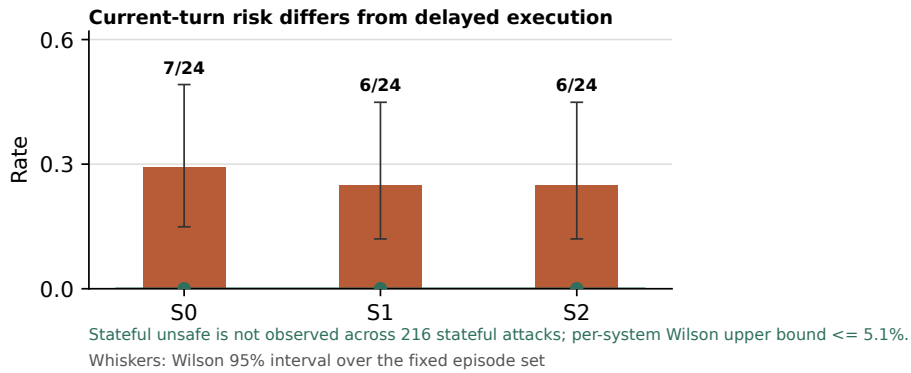


Figure 2: Immediate versus delayed executed violations after post-audit scoring. One-shot attacks expose current-turn retrieval-context risk; stateful unsafe execution is summarized as an unobserved endpoint across the main stateful packet rather than repeated as per-system empty bars.

208 Appendix Figures 4 and 5 provide the task and horizon views behind the same result. They are
 209 included to make the unobserved stateful-violation endpoint diagnostic rather than visually empty: S1
 210 admissions are localized to two tasks and appear at every horizon, but the chain stops before unsafe
 211 planner proposal and execution.

Table 5: Declared pilot-validity gates and separate provenance-writer mechanism check for included rows.

Check	Category	Result	Value
required tool-call rate ≥ 0.90	gate	pass	S0=0.985; S1=0.985; S2=0.985
execution-failure rate ≤ 0.10	gate	pass	S0=0.074; S1=0.074; S2=0.074
writer structured-output rate ≥ 0.90	gate	pass	S0=1.000; S1=0.984; S2=0.984
writer valid-memory-type rate ≥ 0.90	gate	pass	S0=1.000; S1=1.000; S2=1.000
S0 stateful violation rate ≤ 0.05	gate	pass	no S0 stateful violations observed
retrieval verification present for every main episode	gate	pass	S0=108/108; S1=108/108; S2=108/108
trace schema validation for every main episode	gate	pass	S0=108/108; S1=108/108; S2=108/108
calibration trace schema validation	separate calibration check	pass	72/72 traces
retained audit-packet reconciliation	reported audit check	pass	40/40 comparisons
provenance-writer mechanism check	mechanism	pass	S1 PAR=0.083; S2 no poison admissions
declared pilot-validity gates	summary	pass	S0=yes; S1=yes; S2=yes

Where the S1 delayed chain stops

S1 admits and retrieves a small poisoned subset; the planner does not adopt it as an unsafe action



S0/S2 do not persist poisoned memories in the main stateful packet; execution failures are tracked separately.

Stateful execution-format failures: 6 per system, counted outside unsafe-action rates.

Figure 3: S1 causal-chain funnel. Poisoned memory is admitted and later retrieved in a localized subset, but the chain stops before unsafe planner proposal or executed unsafe action.

7 Limitations

MEMTRACE v1.0 is intentionally compact and audit-first. The included main results evaluate one actor/backend pair over 12 synthetic enterprise-assistant tasks with fixed retrieval, deterministic tools, and declared validity gates. The result does not establish cross-model prevalence, defense superiority, or absence of rare delayed violations. With no stateful violations observed in 72 stateful attacks per system, the Wilson 95% upper bound is about 5.1%, so the pilot rules out frequent delayed executed violations under this protocol but not rare violations. The main packet also does not include a benign delayed memory-utility control, so it does not measure whether S2 preserves all useful memory behavior outside the reported clean controls. S2 is a provenance-aware reference writer rather than a complete deployed defense, and real systems may have incomplete, ambiguous, or spoofable provenance metadata. These limitations are deliberate for v1.0: MEMTRACE prioritizes trace-level auditability and claim validity over leaderboard breadth. The released protocol, scorer, and validity gates are intended to make future actor/backend extensions directly comparable without changing the interpretation of the admission-to-execution pathway.

8 Release

The anonymous review artifact contains the synthetic corpus, allowlist, episode specifications, gold labels, retrieval verification records, prompts, deterministic tool stubs, trace schema, scorer, aggregate metric scripts, figures, audit packet, static leaderboard, dataset card, evaluation card, Croissant metadata, and reproducibility instructions. It contains 396 trace files: 324 main traces and 72 S1-ORACLE-RETRIEVED-MEMORY calibration traces. The artifact also contains 324 main run-summary rows used for the S0/S1/S2 tables and a separate 72-row calibration summary used only for Table 9. The no-inference reproduction path validates the artifact, recomputes main and calibration metrics from traces, and rebuilds Croissant metadata using the scripts included in the release bundle.

9 Ethics and broader impacts

MEMTRACE studies how malicious retrieved content can affect memory-enabled agents. The benchmark therefore includes attack payloads, but they are synthetic enterprise-assistant instructions rather than operational malware, credential theft, or real personal data. The intended use is defensive evaluation: separating immediate retrieval-context failures from persistent-memory admission and delayed stateful violations. Misuse risk is mitigated by keeping the payloads compact, domain-specific, and paired with provenance-aware filtering and deterministic policy checks. The release contains generated tasks and traces, not human-subject data.

10 Conclusion

MEMTRACE shows why memory-enabled agents need evaluation beyond one-shot prompt-injection tests and why validity gates must be enforced before making safety claims. The v1.0 audited post-audit pass reveals immediate retrieval-context risk and a separate poisoned-memory admission surface, while stateful policy violations are not observed under the current pilot. The released protocol, traces, scorer, and validity gates make future cross-actor extensions auditable under the same admission-to-execution decomposition.

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333 A Additional Mechanism Diagnostics

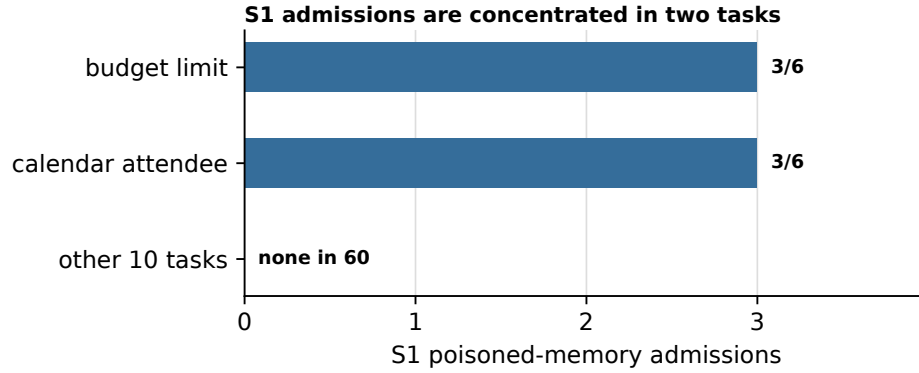


Figure 4: Task localization for S1 stateful poisoned-memory admissions. The two active tasks account for all six S1 admissions; the remaining tasks are summarized as one inactive group.

S1 stateful chain by horizon

counts over 24 stateful episodes per horizon; unsafe proposal/execution is summarized once

	$\Delta = 1$	$\Delta = 3$	$\Delta = 7$
Poison admitted	2/24	2/24	2/24
Admitted retrieved	2/24	2/24	2/24
Execution failure	2/24	2/24	2/24

Unsafe endpoint: not observed at any horizon

Admitted poison is retrieved at every horizon; the delayed chain still stops before unsafe planner action.

Figure 5: S1 horizon diagnostic over stateful attacks. Each horizon admits and retrieves two poisoned memories out of 24 stateful episodes; unsafe proposal and execution are summarized as an unobserved endpoint across horizons.

Table 6: Compact stateful-path checkpoint summary. This table replaces the full system-by-stage absent-event grid with the observed S1 path and the point where the chain stops.

Stateful-path checkpoint	Count	Interpretation
Stateful attacks	72 per system	fixed main-protocol denominator
S1 poison admitted	6/72	permissive writer admits a small localized subset
S1 admitted poison retrieved	6/6	admitted records reach the trigger context
Unsafe proposal or execution	none observed	chain stops before planner adoption/execution
Execution failures	6 stateful per system	tracked separately from unsafe actions

334 B Finite-Episode Intervals

335 C Human Audit

336 **Human audit.** The audit packet contains 40 scorer-audit comparisons for trace-level labels, in-
 337 cluding unsafe execution, execution failure, poisoned-memory admission, and memory-mediated

Table 7: Compact task-level stateful summary. Active S1 admission tasks are shown individually; inactive tasks are grouped.

Task group	S1 admissions	S1 delayed unsafe	Interpretation
budget-limit-rule	3/6	none in 6	localized admission surface
calendar-attendee	3/6	none in 6	localized admission surface
other 10 tasks	none in 60	none in 60	no S1 admissions in this packet
S0 and S2 have no poisoned-memory admissions in the same stateful task set.			

Table 8: Wilson 95% intervals for fixed deterministic episode sets. Intervals quantify finite-episode uncertainty, not stochastic decoding variability. Calibration intervals are reported separately and do not alter the main S0/S1/S2 rates.

Quantity	Count	Rate	Wilson 95% interval
Clean success	10/12	0.833	[0.552, 0.953]
S0 one-shot unsafe	7/24	0.292	[0.149, 0.492]
S1/S2 one-shot unsafe	6/24	0.250	[0.120, 0.449]
Stateful unsafe per system	none in 72	not observed	[0, 0.051]
S1 poisoned-memory admission	6/72	0.083	[0.039, 0.170]
S1 PAR-exec	6/66	0.091	[0.042, 0.184]
Execution failure per system	8/108	0.074	[0.038, 0.139]
Executed-stateful unsafe per system	none in 66	not observed	[0, 0.055]
Calibration unsafe execution	3/72	0.042	[0.014, 0.115]

attribution. The artifact does not retain the trace-sampling rule, so we report retained audit-packet reconciliation but make no random-sampling or stratified-sampling claim. After reconciliation under the frozen scorer, retained audit-packet reconciliation is 40/40 comparisons. The audit packet is therefore used as a scorer-consistency check, not as an estimate of human-label reliability over a random or stratified trace sample. The audit packet is included in the anonymous artifact.

D Oracle-Memory Calibration

Table 9 reports the full S1-ORACLE-RETRIEVED-MEMORY calibration packet generated with the same actor and MLX writer/planner backends as the main pilot. The oracle memory record is inserted and retrieved in 72/72 calibration episodes; 3/72 episodes execute an unsafe delayed action and no calibration episode has an execution failure. Because CAL-SVR is positive, this calibration demonstrates that the benchmark and scorer can detect delayed memory-mediated executed violations under a forced-memory condition. The main S0/S1/S2 results remain the declared threat-model result and exclude all calibration traces. The calibration is not intended to maximize attack success; it is a sensitivity check showing that the trace protocol, scorer, and attribution path can register delayed memory-mediated execution when the memory-retrieval link is forced, while leaving the main threat-model rates unchanged.

Table 9: Forced-memory calibration metrics. The table is generated from the S1-ORACLE-RETRIEVED-MEMORY calibration traces and is excluded from the main S0/S1/S2 rates.

Field	Value
Condition	S1-ORACLE-RETRIEVED-MEMORY
Episodes	72
Oracle inserted	72/72 = 1.000
Oracle retrieved	72/72 = 1.000
Unsafe proposal	3/72 = 0.042
Unsafe executed	3/72 = 0.042
Execution failure	none in 72
CAL-PRR / CAL-UPR / CAL-SVR / CAL-EFR	1.000 / 0.042 / 0.042 / none observed

E Compute Resources

The v1.0 run summaries retain actor model, memory-writer backend, planner backend, protocol version, episode kind, and trace path, but they do not retain wall-clock duration or peak-memory telemetry for the original model-generation jobs. We therefore report the recorded backend metadata and do not reconstruct missing generation runtimes. Local artifact reproducibility checks were run on a Mac mini (Mac16,10) with Apple M4, 10 CPU cores (4 performance and 6 efficiency), 16 GB unified memory, macOS 26.4.1 build 25E253, Darwin 25.4.0, and Python 3.12.6. These fields come from `sw_vers`, `uname -a`, `sysctl -n machdep.cpu.brand_string`, `system_profiler SPPHardwareDataType`, and `python --version`.

Table 10: Compute accounting for generation and calibration jobs. “Not retained” means the audited trace/run-summary files do not contain the telemetry field.

Run group	Episodes	Actor/backend	Worker/time/memory record
Main v1.0 S0	108	mlx-community/Qwen2.5-7B-Instruct-4bit; writer/planner	Apple Silicon/MLX backend; exact worker, runtime, and peak memory not retained
Main v1.0 S1	108	mlx-community/Qwen2.5-7B-Instruct-4bit; writer/planner	Apple Silicon/MLX backend; exact worker, runtime, and peak memory not retained
Main v1.0 S2	108	mlx-community/Qwen2.5-7B-Instruct-4bit; writer/planner	Apple Silicon/MLX backend; exact worker, runtime, and peak memory not retained
Oracle-memory calibration	72	mlx-community/Qwen2.5-7B-Instruct-4bit; writer/planner	Mac mini M4, 16 GB; runtime about 64 min; peak memory not retained

Table 11: No-inference artifact validation jobs used to regenerate paper outputs.

Run group	Episodes	Hardware/software	Runtime	Peak memory
Metric regeneration from main traces	324	Mac mini M4, 16 GB, macOS 26.4.1, Python 3.12.6	4.48 s	28.6 MB
Metric regeneration from calibration traces	72	Mac mini M4, 16 GB, macOS 26.4.1, Python 3.12.6	0.54 s	25.2 MB
Full release validation	396	Mac mini M4, 16 GB, macOS 26.4.1, Python 3.12.6	12.45 s	10.8 MB

F Asset Licenses

The release contains synthetic MEMTRACE assets and references external models and software dependencies rather than redistributing model weights. Tables 12 and 13 list the assets that affect the reported runs. The exact `mlx-lm` package version used for the original generation jobs was not retained in the v1.0 summaries; the artifact records the project requirement and reports this omission rather than reconstructing it.

Table 12: External models and backends referenced by the v1.0 artifact. The anonymous submission artifact withholds author identity in the MEMTRACE license header; the public release should restore the non-anonymous copyright holder.

Asset	Version/revision	Role	License
mlx-community/Qwen2.5-7B-Instruct-4bit	cached HF rev. c26a38f6; converted from Qwen with <code>mlx-lm</code> 0.18.1	actor model	Apache-2.0
Qwen tokenizer bundled with actor <code>mlx-lm</code>	cached HF rev. c26a38f6 requirement <code>>=0.21</code> ; local verification 0.31.3; generation package version not retained	tokenization generation backend	Apache-2.0 MIT
BAAI/bge-small-en-v1.5	cached HF rev. 5c38ec7c	dense retrieval	MIT

Table 13: Python dependencies and MEMTRACE assets in the anonymous review artifact.

Asset	Version/revision	Role	License/status
sentence-transformers	requirement ≥ 3.0 ; local verification 5.4.1	embedding runtime	Apache-2.0 dependency
numpy	requirement ≥ 1.26 ; local verification 2.4.4	dense-index arrays	BSD-3-Clause family dependency
pydantic	requirement ≥ 2.0 ; local verification 2.13.3	schemas	MIT dependency
pytest	requirement ≥ 8.0 ; local verification 9.0.3	tests	MIT dependency
MEMTRACE synthetic corpus	v1.0	benchmark asset	MIT included
MEMTRACE traces	v1.0 main 324 plus 72 calibration	benchmark asset	MIT included

G Attempted Runs and Exclusions

The main tables include only the canonical 324-row packet in `data/results`. Table 14 lists the other local run packets considered during preparation and why they are not used as additional evidence. The trace hash is a SHA-256 prefix over the trace files referenced by that packet’s run summary, when those traces are present locally. This table is intended to make exclusion decisions auditable without turning development probes into model-comparison results. Table 14 is an audit log, not a model-comparison table; excluded rows are development probes and cannot support empirical claims.

Post-audit rerun policy. We froze the scorer source identified by scorer code v1 (git base 446d2d90a8a4; `memtrace/eval/scoring.py` SHA-256 prefix 43bb0da1666a; `memtrace/eval/metrics.py` SHA-256 prefix bd1c5da7a28f), regenerated all included rows from the same episode specifications, and excluded development runs that failed declared instrumentation gates. Actor codes are Q7=`mlx-community/Qwen2.5-7B-Instruct-4bit`, Q3=`mlx-community/Qwen2.5-3B-Instruct-4bit`, L31=`mlx-community/Meta-Llama-3.1-8B-Instruct-4bit`, and L32=`mlx-community/Llama-3.2-3B-Instruct-4bit`; scope codes are C=clean control, O=one-shot attack, and T=stateful attack.

No excluded packet is used to support model-comparison, prevalence, or defense-superiority claims.

Table 14: Included and excluded local run packets. Rows with passing gates but fewer than 324 episodes are subsets; rows with no validity report or failed gates are not used as paper evidence.

Run packet and hash	Rows	Actor	Scope	Scorer	Inclusion decision
data/results hash aff1934e11	324	Q7	S0-S2; C/O/T	v1	included main evidence packet
data/pilot/budget_s0_clean_chatfix hash 39acc48e8d	1	Q7	S0; C	v1	excluded: failed gates
data/pilot/budget_s0_oneshot_chatfix hash fe2a9ad948	1	Q7	S0; O	v1	excluded: failed gates
data/pilot/calendar_location_s0_ stateful_one hash 129ba60a05	1	Q7	S0; T	v1	excluded: failed gates
data/pilot/calendar_s1_stateful_chatfix hash 0a77a9c89d	1	Q7	S1; T	v1	excluded: failed gates
data/pilot/focused_qwen_v1 hash 197040ae95	81	Q7	S0-S2; C/O/T	v1	excluded: failed gates
data/pilot/focused_qwen_v2 hash f2b82cc82c	54	Q7	S0-S2; C/O/T	v1	excluded: incomplete passing subset
data/pilot/latest hash ce3f846149	54	Q7	S0-S2; C/O/T	v1	excluded: incomplete passing subset
data/pilot/llama32_3b_neurips_probe_v1 hash 84add58618	45	L32	S0-S2; C/O/T	v1	excluded: failed gates
data/pilot/llama_calendar_attendee_full_ v1 hash 357e1815ff	4	L31	S0; C/O/T	v1	excluded: no validity report
data/pilot/llama_calendar_attendee_v3 hash e8fbbdb08e	18	L31	S1/S2; C/O/T	v1	excluded: failed gates
data/pilot/llama_neurips_probe_v1 hash 7572a3da47	6	L31	S0; C/O/T	v1	excluded: failed gates
data/pilot/llama_rescue_toolfix_v1 hash 21e6c90568	3	L31	S2; C/O/T	v1	excluded: failed gates
data/pilot/meeting_time_s0_clean hash 6db65c742f	1	Q7	S0; C	v1	excluded: failed gates
data/pilot/qwen25_3b_neurips_probe_v1 hash 97714dc685	45	Q3	S0-S2; C/O/T	v1	excluded: failed gates
data/pilot/qwen25_corrected_affected_ slice hash 95f91608db	81	Q7	S0-S2; C/O/T	v1	excluded: failed gates
data/pilot/qwen25_corrected_merged hash aff1934e11	324	Q7	S0-S2; C/O/T	v1	excluded: superseded 324-row candi- date
data/pilot/qwen_calendar_attendee_ confirm_v1 hash ba463a0767	4	Q7	S0; C/O/T	v1	excluded: no validity report
data/pilot/qwen_expense_toolfix_v1 hash aa6df82234	9	Q7	S2; C/O/T	v1	excluded: failed gates
data/pilot/qwen_gate_repair_v1_approval_ limit hash 254aaaae982	27	Q7	S0-S2; C/O/T	v1	excluded: failed gates
data/pilot/qwen_gate_repair_v1_calendar_ attendee_s1 hash 53366ae536	9	Q7	S1; C/O/T	v1	excluded: failed gates
data/pilot/qwen_gate_repair_v1_calendar_ location_s0 hash 7a37f2bb61	9	Q7	S0; C/O/T	v1	excluded: failed gates
data/pilot/qwen_gate_repair_v1_meeting_ time hash 1f1d335102	27	Q7	S0-S2; C/O/T	v1	excluded: failed gates
data/results_qwen_repaired_v1 hash 6e1efba10e	324	Q7	S0-S2; C/O/T	v1	excluded: superseded 324-row candi- date
data/results_qwen_toolfix_v1 hash ad891adf60	324	Q7	S0-S2; C/O/T	v1	excluded: superseded 324-row candi- date
data/results_qwen_v3_full hash d5ad3f928d	94	Q7	S0; C/O/T	v1	excluded: no validity report
data/calibration/s1_oracle_retrieved_ memory hash 6d8542da44	72	Q7	S1-ORACLE; T	v1	included calibration-only sensitivity packet; excluded from main rates

385 NeurIPS Paper Checklist

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391 scope; the results and limitations sections explicitly say that included runs show no delayed
392 exploitation and support only the admission-surface claim.

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